

Exploring the Role of Loss Functions in Deepfake Detection

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Abstract—With deepfakes becoming more common, it's increasingly difficult to separate real content from manipulated ones. As vast amounts of data are generated daily, the need for effective deepfake detection becomes increasingly critical. We aim to apply a basic CNN model to check if an image is real or fake. We examine how various loss functions, including binary cross-entropy, focal loss, and hinge loss, impact training and detection performance. The models were evaluated on a dataset of 190,335 images with a focus on accuracy and loss. Binary Cross-Entropy performs well on balanced datasets, Focal Loss shows superior performance in imbalanced datasets, and Hinge Loss provides robust margin-based classification. Our findings provide insights into the optimal loss functions for enhancing the performance of deepfake detection models. At 92.70% training accuracy and a 0.0118 training loss, the Focal Loss model emerged as the best, providing a solid foundation for future advancements in deepfake detection.

Keywords—CNN, GAN, AI, Deepfakes

I. INTRODUCTION

In recent technological development, the rapid growth of AI has given rise to many techniques for digitally manipulating images. These manipulated images are referred as Deepfakes. Deepfake is a term for fake images, audios and videos generated using AI. These can be almost identical to genuine images, which raises worries about how this technology might be misused. Making a deepfake might sound very complicated but it is a very easy process. Figure 1 shows some fake images which are almost similar to genuine images and very hard to interpret by human eyes. The tools for creating deepfakes are available to everyone which makes the deepfake a potential threat for the society. Deepfakes are generally created using two techniques. One example is Autoencoders, a neural network used for unsupervised learning that works through Encoding and Decoding. A different strategy is employing Generative Adversarial Networks (GANs), where the generator creates fake content from random noise, and the discriminator judges whether it's real or fake [1].

A. Ethical Implications of Deepfakes

The ability to create convincing deepfakes has far reaching implications, particularly in areas such as misinformation, privacy violations, and cybersecurity [2]. This section explores the ethical implications of deepfakes, legal challenges, and their quantifiable impact on society. Deepfakes weaken our trust in digital media by making fake content look real [10]. Politically, deepfakes can generate fake interviews or speeches from politicians, possibly

affecting election outcomes. With over 900 million eligible voters, elections in India are a massive effort. According to a 2019 study, nearly 70% of Indians find it hard to recognize fake news online [18]. So far, deepfakes haven't impacted Indian elections, but experts think it's just a matter of time. India's legal system doesn't yet have deepfake-specific rules, but laws such as IT Act 2000, IT Rules 2021, and some BNS 2023 sections address AI-generated misinformation [19]. India's legal system struggles with deepfakes because there is no law specifically for them. To confirm if digital content is real, governments should support AI-based verification tools. Strong AI detection tools, such as Microsoft's Video Authenticator and Google's DFDC, along with media literacy programs, can help fight deepfakes. Social media platforms could be legally required to include deepfake detection tools. Some sample fake images used for model training can be seen in Figure 1.



Fig. 1 Sample Fake Images from the Dataset Used for Model Training

Table 1 outlines key aspects related to the ethical and societal concerns of deepfakes. It highlights the issue of privacy, lack of consent, legal challenges, and the broader social impact. It explains how deepfake technology can affect individuals and society by undermining trust and raising ethical questions.

TABLE I. ETHICAL IMPLICATIONS OF DEEPFAKES

ASPECT	DESCRIPTION
Privacy	Concerns about unauthorized use of personal images and videos.
Consent	Making and sharing deepfakes without the subject's permission is a serious ethical issue.
Legal	Deepfake technology is advancing so quickly that laws and regulations are struggling to catch up.
Social Impact	Deepfakes can erode public trust in media and digital content

B. Deepfake Tools

With deep learning technology improving, deepfake tools have made it easier to alter media convincingly. Table 2 provides a comparison of popular tools used in the creation and manipulation of deepfakes. It includes open-source software for face swapping and reenactment, highlighting each tool's purpose and its corresponding GitHub repository for further exploration.

TABLE II. COMPARATIVE STUDY OF DEEPAKE TOOLS

Feature	FaceSwap	DeepFace Lab	FSGAN	StyleRig	MarioNET
Links	FaceSwap Github	DeepFace Lab Github	FSGAN Github	StyleRig: 3D Control	MarioNET e Github
Core Technology	Autoencoders, and GANs	Autoencoders, GANs, and XSeg [20]	Few-Shot GANs and encoder-decoder networks [21]	Deep learning-based facial rigging, pose and expression control	Neural rendering and rigging [4]
Primary Use Case	Open-source access for face swapping in images and videos.	High-quality deepfake generation and video manipulation.	Real-time face manipulation in video with minimal training data.	Detailed control over facial expressions, animations, and head poses [3]	Realistic 3D avatar manipulation with facial reenactment and animation
Output Quality	High-quality results in face-swapping, particularly with image data	Very high, producing near-photorealistic results in video face swaps with precise facial alignment	High-quality, but less realistic compared to DeepFaceLab for high-quality video swaps	High-quality expression manipulations with fine-tuned control over poses and expressions	Very high, especially in 3D modeling and facial animations, creating lifelike avatars

The study investigates how changes in loss functions affect the efficiency of deepfake detection models. Specifically, we investigate three loss functions: Binary-Cross Entropy Loss, Focal Loss, and Hinge Loss.

II. LITERATURE REVIEW

This literature review surveys existing strategies for deepfake detection, focusing on their performance and limitations.

A. Convolutional Neural Networks based Techniques

CNNs are a preferred method for identifying deepfakes, thanks to their skill in extracting detailed visual features. The study in [11] highlighted how pre-trained models like ResNet and VGG were adapted to specific datasets to identify signs of tampering. According to [13], CNNs are capable of detecting subtle inconsistencies in facial details and textures, delivering high accuracy on standard datasets. Both studies highlighted challenges in adapting these models to emerging types of deepfakes, particularly as generation methods advance.

B. Temporal and Spatiotemporal Approaches

Temporal artifacts, such as jerky head movements or persistent flickering, are typical indicators of deepfake videos. The study in [12] focused on utilizing RNNs and LSTMs to analyze frame-by-frame sequences, leading to improved detection capabilities. The spatiotemporal

approach in [14] combines the image analysis strengths of CNNs with the temporal processing power of RNNs for effective detection. These methods, although effective, face difficulties with computational costs and scalability when applied to real-time detection.

C. Frequency Domain Analysis

While deepfake artifacts are difficult to spot in the spatial domain, they are more apparent in the frequency domain. [9] studied how Discrete Fourier Transform (DFT) and Wavelet Transform can be used to spot compression artifacts and blurring in deepfakes. This strategy proved useful for detecting lower-quality deepfakes but had limitations with high-resolution outputs, as described in [8].

D. Adversarial and Ensemble Learning

Adversarial learning techniques have been suggested to counter the complexity of GAN-generated deepfakes. [15] detailed how adversarial strategies include the parallel training of detection models and GANs, fostering a dynamic adversarial atmosphere. Additionally, [7] emphasized the use of ensemble learning methods that merge predictions from various models to improve robustness, especially in detecting different types of deepfakes.

E. Explainable AI in Deepfake Detection

The goal of Explainable AI (XAI) methods is to enhance the clarity as well as transparency in detecting deepfakes. [6] described how saliency maps and Grad-CAM help visualize which input areas contribute to the model's classification. This enhances the model's interpretability and highlights areas of vulnerability, as detailed in [16]. However, the extra computational cost still poses a major disadvantage.

III. METHODOLOGY

The methodology explores the impact of different loss functions in deepfake detection through Convolutional Neural Networks (CNNs). This section offers a detailed overview of the entire process, starting with data preprocessing and augmentation to ensure high-quality model training. The CNN model is designed to identify key features for deepfake detection. The following subsections detail each component of the methodology, providing insights into the experimental setup:

A. Dataset Used

Our models were trained with a dataset that contained 190,335 images. The dataset was collected from kaggle, which is named as "deepfake and real images" [5]. It contains digitally manipulated and real images of human faces which are created by various means. Originally available at OpenForensics under the title 'Multi-Face Forgery Detection And Segmentation In-The-Wild Dataset [V.1.0.0]' (Zenodo), the dataset was meticulously processed to optimize the input for our models [17].

The deepfake images dataset was divided into three parts: Test, Train, and Validation. The Training set includes 140,002 images, split evenly between 70,001 fake and 70,001 real images to maintain class balance. The Validation set includes 39,428 images, with 19,641 fake and 19,787 real images, and the Test set comprises 10,905 images, with 5,492 fake and 5,413 real images.

B. Image Preprocessing, Augmentation, and Model Compilation

In this project, image data preprocessing and augmentation are handled using Keras' 'ImageDataGenerator' class. This process helps to improve model generalization and performance by introducing variability and preprocessing the image data. The test data is primarily rescaled to ensure consistency in preprocessing. The images are processed by scaling their pixel values from '[0, 255]' to '[0, 1]'. Normalizing the pixel values helps the model train faster and learn more effectively. A random shearing transformation is applied to the images. Shearing introduces a tilt to the images, with a shear intensity of 20%. This augmentation helps the model adapt to different scales. The images undergo random zooming within a range of 20%. This augmentation helps the model to handle variations in scale. Images are randomly flipped horizontally. This augmentation introduces variations in orientation, which helps in making the model constant to horizontal flips. Each batch processes 32 images. The batch size controls how many images are processed to adjust the model weights at each step. The classification type is set to binary, so that the model will perform binary classification. The labels for the images will be binary (0 or 1), corresponding to the two classes.

The rescaling, shearing, zooming, and flipping augmentations enhance the diversity of the data which is being used for training, while the resizing, batch size, and class mode configurations ensures appropriate preparedness of data for training a binary classification model. It helps the model generalize better and perform effectively on unfamiliar data.

For model compilation, the Adam optimizer was chosen. The Adam optimizer ensures efficient convergence during training by dynamically adjusting learning rates and the Accuracy Metric is used to monitor performance during training.

C. Model Architecture

CNNs are deep learning models made for handling images and recognizing patterns. Instead of relying on manual features like traditional machine learning, CNNs learn image representations through multiple convolution and pooling layers. This makes CNNs especially powerful for recognizing objects, studying faces, and spotting deepfakes. For this project, three CNN-based models were created, each trained with a different loss function—Binary Cross Entropy (BCE), Focal Loss, and Hinge Loss—to assess their effectiveness in classifying genuine and manipulated images. These loss functions were picked because each one impacts the model's learning and results differently. BCE is widely used for binary classification as it checks how well predicted probabilities match actual labels, making it a standard for deepfake detection. An improved version of BCE, Focal Loss focuses on challenging cases to handle class imbalance. The reason for choosing this function was to measure its effect on enhancing model flexibility. Hinge Loss was used to check if a wider separation between real & fake images could boost detection accuracy, compared to probabilistic methods like BCE and Focal Loss.

1. Binary Cross Entropy Loss Model:

The architecture of the BCE Loss model is tailored to classify genuine and fake images, using the BCE function to enhance binary classification accuracy. The model uses a 2D

convolutional layer with 32 filters and a 3×3 kernel size, along with ReLU activation, to capture spatial features from the input image. Resized RGB images are used as input, with dimensions of $64 \times 64 \times 3$. A 2×2 max pooling layer helps decrease computational load by reducing feature map size while maintaining crucial details. A second 3×3 convolutional layer with 32 filters is implemented, with ReLU activation to enhance non-linearity. A max-pooling operation is applied once more to decrease the dimensionality. Before passing to the fully connected layers, the 2D feature maps are transformed into a 1D vector. A fully connected layer containing 128 neurons with ReLU activation is used to capture intricate patterns. The output layer has one neuron with a sigmoid activation, producing a probability for binary classification. Training was limited to 10 epochs, while early stopping and checkpoints helped monitor progress and save the top model. If validation loss failed to improve for three epochs, early stopping stopped training and loaded the best weights. Model checkpointing preserved the model that achieved the lowest validation loss to maintain optimal performance.

2. Focal Loss Model:

The architectural design was consistent with the BCE model, with modifications to the loss function. Focal loss is defined mathematically as,

$$FL(y_{true}, y_{pred}) = -\alpha(1 - p_t)^\gamma \cdot \log(p_t) \quad (1)$$

Where, the predicted probability for the correct class is represented by p_t , α is a parameter used to balance the class distribution, and γ helps to lower the loss from examples that are already correctly classified.

For this project, the focal loss was implemented with $\gamma = 2.0$ and $\alpha = 0.25$ to ensure that the model focuses on difficult examples during training to make the model more robust. The model was trained up to 10 epochs, with validation data used to monitor performance.

3. Hinge Loss Model:

Like the other loss models, the Hinge Loss model follows the same architecture and preprocessing steps, but its loss function adds a unique optimization mechanism. For training, the model utilized hinge loss, a common choice for binary classification problems with -1 and +1 as the labels. The loss function is mathematically defined as;

$$(y_{true}, y_{pred}) = \text{mean}(\max(0, 1 - y_{true} \cdot y_{pred})) \quad (2)$$

Where, y_{true} is the true label i.e., -1 or +1 and y_{pred} is the value predicted by the model's linear output layer.

The model was trained over 10 epochs with a validation set to assess generalization. Early stopping and checkpointing were not utilized in this configuration.

D. Training and Validation Metrics

The models are tested based on their accuracy and loss during training and validation over ten epochs. They provide a clear picture of how the models learn and generalize to unknown cases. Accuracy measures how many samples were classified correctly out of the total. Training accuracy indicates how accurately the model learned from the training data. If the training accuracy is high, the model is picking up patterns correctly from the dataset. Validation accuracy

indicates how well the model works on unseen data. This is a key measure to see if the model can generalize or is too focused on training data.

Loss measures the gap between predicted and real result. Training loss measures how accurately the model makes predictions on the training dataset. If training loss is low, the model's predictions are more accurate. By computing validation loss, we can see whether the model's performance is improving or deteriorating. When validation and training loss drop together, it indicates effective learning. An increasing validation loss with a decreasing training loss is a sign that the model is overfitting.

We observed the loss values over time to confirm that the model was learning efficiently. The optimization process relies on loss functions to help the model learn effectively. Tracking these metrics allowed us to see if the models were learning correctly and fix any fitting issues..

IV. RESULTS

This section provides a comparison of the models' performance:

1. Binary Cross Entropy Loss Model

In the first epoch, a training accuracy of 72.90% with a loss of 0.5231 was achieved. The validation accuracy was 78.79% with a loss of 0.4453. By the fifth epoch, accuracy improved to 89.79% with a lower loss of 0.2378, and the validation accuracy rose to 86.03% with a loss of 0.3248. By the tenth epoch, the final accuracy reached 91.28%, with a training loss of 0.2087, while the validation accuracy was 85.28% with a loss of 0.3401. Figure 2 shows results of BCE model's prediction of 9 random images from the test data.



Fig. 2 Prediction of 9 random images from the Test Set by Binary Cross Entropy Model

2. Focal Loss Model

In the first epoch, a high accuracy of 89.87% with a very low loss of 0.0163 on the training set was observed while, the validation accuracy was 83.70% with a loss of 0.0230. By the fifth epoch, accuracy improved to 91.79% with a training loss of 0.0131, and the validation accuracy was 85.67% with a loss of 0.0232. By the tenth epoch, the final accuracy reached

92.70%, with a training loss of 0.0118, while the validation accuracy was 85.94% with a loss of 0.0234. Figure 3 shows how the Focal Loss Model predicted 9 random test images.



Fig. 3 Prediction of 9 random images from the Test Set by Focal Loss Model

3. Hinge Loss Model

In the first epoch, the model achieved a high accuracy of 91.85% with a loss of 0.5820 on the training set. The validation accuracy was 85.08% with a loss of 0.6530. By the fifth epoch, accuracy slightly increased to 91.70%, with a loss of 0.5824, and the validation accuracy reached 86.97% with a loss of 0.6340. By the tenth epoch, the final accuracy was 91.58% with a training loss of 0.5837, while the validation accuracy was 85.90% with a loss of 0.6446. Prediction of 9 random test images by Hinge Loss Model are shown in Figure 4.



Fig. 4 Prediction of 9 random images from the Test Set by Hinge Loss Model

Table 3 provides a summary comparative analysis for all the three models.

TABLE III. SUMMARY OF COMPARATIVE ANALYSIS

Metric	Binary Cross Entropy	Focal Loss	Hinge Loss
Training Accuracy (Final)	91.28%	92.70%	91.58%
Validation Accuracy (Final)	85.28%	85.94%	85.90%
Training Loss (Final)	0.2087	0.0118	0.5837
Validation Loss (Final)	0.3401	0.0234	0.6446
Convergence Speed	Moderate	Fast	Slow
Generalization	Good	Good	Moderate
Overfitting Tendency	Low-to-Moderate	Moderate	Moderate

Figure 5, 6, and 7 display the accuracy and loss graphs for BCE, Focal Loss, and Hinge Loss respectively.

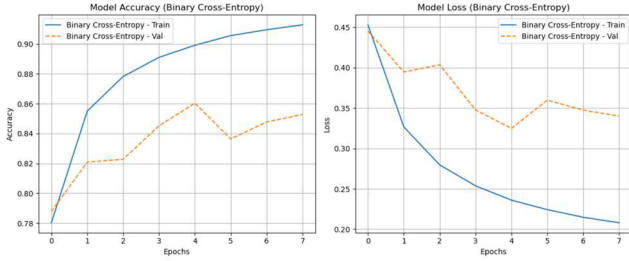


Fig. 5 Accuracy and Loss curves for Binary Cross Entropy Loss

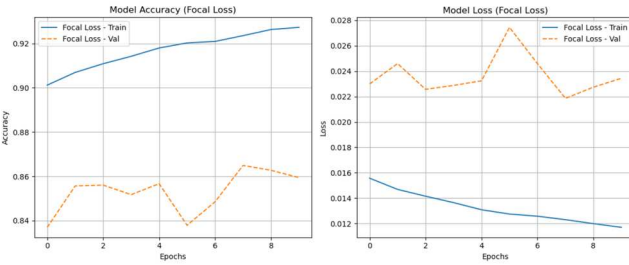


Fig. 6 Accuracy and Loss curves for Focal Loss

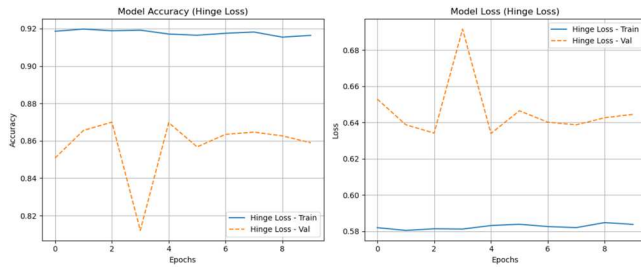


Fig. 7 Accuracy and Loss curves for Hinge Loss

BCE model showed steady improvement with balanced accuracy and loss metrics, making it a reliable choice for this task. It performed well without significant overfitting, though it required careful monitoring during training. Focal Loss model quickly reached high accuracy and low loss values, indicating strong confidence in predictions, particularly in cases with class imbalance. However, the risk of overfitting was higher, and validation performance plateaued earlier. While the Hinge Loss model maintained high training accuracy, it struggled with higher validation loss, suggesting

less effective generalization compared to the other two models. It may require more tuning or different architecture adjustments to perform on par with the others. Figure 8 compares the accuracy and loss curves for all three loss functions.

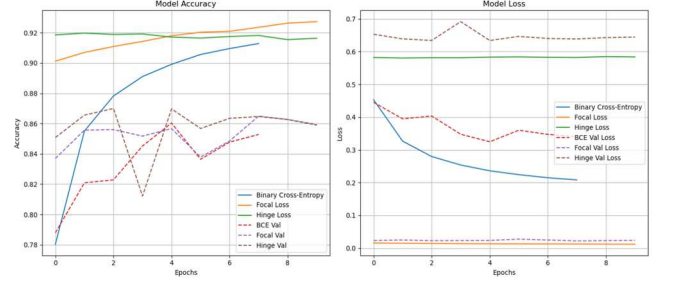


Fig. 8 Accuracy and Loss curves comparison between Binary Cross Entropy Loss, Focal Loss, and Hinge Loss

V. DISCUSSION

A. Implications of Findings

This analysis offers key takeaways for the design and optimization of deepfake detection systems. The BCE model converges relatively well, with steady improvement in both training and validation accuracy across epochs. Overfitting became noticeable after the 5th epoch, with the validation loss ceasing to improve, while the training accuracy continued to grow. With 91.28% training accuracy and 85.28% validation accuracy, the model performed well on unseen data.

The Focal Loss model showed very high training accuracy from the beginning, suggesting effective handling of the dataset by focusing on hard examples. Despite having minimal training loss, the validation accuracy plateaued at 85-86%, hinting at slight overfitting, even with focal loss. Slightly higher training accuracy compared to the BCE model, but similar validation accuracy. The low loss values indicate strong confidence in predictions.

The Hinge Loss model converges relatively slower compared to the other two, as indicated by the marginal changes in accuracy and loss across epochs. While the model consistently achieved high training accuracy, the validation loss remained relatively high, indicating potential issues with generalization. It achieved good training accuracy but struggled with validation loss, which stayed high throughout, suggesting that the Hinge Loss model might not be as effective for this specific task.

B. Real-World Applications

By applying Focal Loss, detection systems could achieve higher accuracy rates, especially in environments where false negatives (failing to detect a deepfake) carry significant consequences, such as in media verification or legal evidence. Since Focal Loss keeps performance steady across epochs, the model can handle new deepfakes better, improving real-world reliability.

C. Acknowledgement of Limitations

The results are specific to the CNN architecture used. Other architectures, such as Vision Transformers, EfficientNet, or ResNet, might interact differently with the loss functions testes and may yield better results. The number

of epochs and the training duration may have been limited by available computational resources, potentially affecting the thoroughness of the model's training and evaluation. While this study focused on loss functions, other hyperparameters (e.g., learning rate, batch size) and techniques (e.g., data augmentation, regularization) were not varied, which could also influence the outcomes.

Expanding the dataset, including deepfakes generated by emerging techniques, could provide a more rigorous test of model robustness. This could involve crowd-sourced datasets or collaboration with platforms like social media companies that encounter a wide array of deepfake content. Future research could explore other specialized loss functions, such as Dice Loss, Jaccard Loss, or hybrid approaches combining multiple loss functions, to further optimize deepfake detection.

VI. CONCLUSION

In this research, the efficiency of three distinct loss functions—Binary Cross Entropy Loss, Focal Loss, and Hinge Loss—within Convolutional Neural Network (CNN) was explored. We trained and evaluated each model using a diverse set of real and altered images. The aim was to compare the performance of these loss functions based on accuracy, precision, recall, and F1-score.

In conclusion, this research contributes in finding the most effective deepfake detector. Future research could explore hybrid models that integrate the advantages of various loss functions to improve detection accuracy.

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